

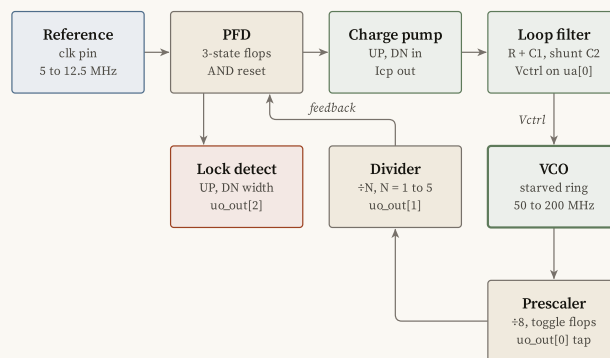
The charge-pump PLL, explained

A phase-locked loop makes a fast clock from a slow one by steering an oscillator until a divided copy of it lines up with the reference.

Ours multiplies a 5 to 12.5 MHz reference on `clk` by $8N$, with $N = 1$ to 5 on the input pins. The phase-frequency detector (PFD) compares the reference edge with the edge returning from the dividers. A late feedback edge means the voltage-controlled oscillator (VCO) is slow, so the PFD raises UP for the gap between the edges; an early one raises DN. The charge pump turns those pulses into $+I_{cp}$ or $-I_{cp}$, a packet of charge proportional to the phase error, and the loop filter integrates it into V_{ctrl} , which sets the VCO frequency. The VCO output goes through $\div 8$ and $\div N$ back to the PFD.

At lock the two edges line up, UP and DN shrink to tiny equal pulses, V_{ctrl} holds and $f_{vco} = 8Nf_{ref}$. Because the filter integrates, any steady error keeps adding charge until it is gone. The lock detector raises its pin once UP and DN stay short.

FIGURE 1 *the loop, from the reference back to the PFD*



Slate is the reference pin, sand is standard-cell logic tested in CocoTB, sage is analog simulated in ngspice across corners, and rust is the lock check. The proposal's system diagram, [figures/cppll-blocks.json](#).

Why each number is what it is

The $\div 8$ prescaler. Synthesized logic is estimated to run at 50 MHz at most, and the VCO reaches about 333 MHz at the fast corner (EST.). Three toggle flops divide it first: $333/2 = 166$, $333/4 = 83$, $333/8 = 41.6$ MHz. A $\div 4$ makes exactly 50 MHz at the typical 200 MHz but fails at the fast corner, so $\div 8$ is the smallest power of two that works.

The reference range and N . The loop locks only where $8Nf_{ref}$ falls in the VCO's typical 50 to 200 MHz. With $N = 1$ to 5 every reference from 5 to 12.5 MHz has a lock point: $N = 2$ to 5 at 5 MHz, 1 to 4 at the nominal 6.25 MHz ($N = 2$ gives 100 MHz), 1 and 2 at 10 and 12.5 MHz.

The loop filter. Gardner's analysis gives the natural frequency and damping:

$$\omega_n = \sqrt{\frac{I_{cp} K_{vco}}{2\pi PN C_1}}, \quad \zeta = \frac{R C_1 \omega_n}{2}, \quad (1)$$

with K_{vco} in rad/s per volt, PN the total divide ratio ($P = 8$) and R, C_1 the filter's series pair. With $I_{cp} = 20 \mu\text{A}$, $K_{vco} = 2\pi \cdot 100 \text{ MHz/V}$ (EST.), $PN = 16$ and $C_1 = 20 \text{ pF}$, $\omega_n = 2.5 \times 10^6 \text{ rad/s}$, about $2\pi \cdot 400 \text{ kHz}$. The PFD samples once per reference cycle, so the bandwidth must stay well under f_{ref} : at 6.25 MHz the 490 kHz crossover is $f_{ref}/13$, under the $f_{ref}/10$ rule.

For $\zeta \approx 0.55$, $R = 2\zeta/(C_1\omega_n) \approx 22 \text{ k}\Omega$, the written proposal's hand value; the slide's $26 \text{ k}\Omega$ gives $\zeta \approx 0.65$. $C_2 = C_1/10$ smooths the ripple each pump pulse leaves. With $22 \text{ k}\Omega$ (zero 360 kHz, C_2 pole 4 MHz) the hand phase margin is about 47° (EST.), 51° with $26 \text{ k}\Omega$. The first typical-corner ngspice run measured 49.5° at $8N = 16$, above the 45° target. Both ω_n and ζ scale as $\sqrt{I_{cp}/PN}$, so ζ falls from 0.78 at $PN = 8$ to 0.35 at $PN = 40$.

Lock time. Settling takes about $4/(\zeta\omega_n) = 4/(0.55 \times 2.5 \times 10^6) \approx 3 \mu\text{s}$ at $PN = 16$ and $7 \mu\text{s}$ at $PN = 40$ (EST.). Acquisition adds about $2 \mu\text{s}$: V_{ctrl} slews at $I_{cp}/(C_1 + C_2) \approx 0.9 \text{ V}/\mu\text{s}$ over a 1.5 V swing.

The capacitor. At an assumed MIM density of 1 to 2 fF/ μm^2 , 20 pF needs 10,000 to 20,000 μm^2 (EST.), a square about 100 to 140 μm on a side and a large share of a two-tile analog area. If it does not fit, the node between R and C_1 goes out on a third analog pin and C_1 moves to the board.

FIX BEFORE PRESENTING

The slide's $26 \text{ k}\Omega$ gives $\zeta \approx 0.65$; the 0.55 and 47° are for $22 \text{ k}\Omega$. Pick one and quote its own margin.

The pins and the plan

Every pin either sets the loop up or lets you see inside it; nothing faster than tens of MHz gets through the pads.

TABLE 1 *every pin of the analog tile, and why it is there*

PIN	DIR	USE, AND WHY
clk	in	The reference, 5 to 12.5 MHz, which the PFD compares against.
rst_n, ena	in	Tiny Tapeout's standard reset and enable.
ui_in[2:0]	in	N , 1 to 5 (three bits; codes 0, 6 and 7 unsupported).
ui_in[6]	in	PLL enable: powers down the pump and VCO.
ui_in[7]	in	Picks $\div 8$ or $\div 16$ on <code>uo_out[0]</code> , for a slow pad or scope.
uo_out[0]	out	VCO $\div 8$ or $\div 16$, 3.1 to 25 MHz: pads cannot carry 200 MHz.
uo_out[1]	out	Feedback divider output. At lock it equals f_{ref} , and <code>uo_out[0]</code> in $\div 8$ mode is $N f_{ref}$, so their ratio reads N .
uo_out[2]	out	Lock flag.
uo_out[4:3]	out	UP and DN for debug; short pulses may not survive the pads.
uio[1:0]	in	Pump current trim; doubles I_{cp} at $PN = 32$ to 40 for damping.
ua[0]	analog	V_{ctrl} , the filter node. Read through a buffer or with the loop open: a 15 pF scope probe would push the margin under 20°.
ua[1]	analog	Bias: an external resistor sets the pump's mirror (optional).
Spare: <code>ui_in[5:3]</code> and <code>uio[7:2]</code> as inputs; <code>uo_out[7:5]</code> driven low.		

TABLE 2 *the milestones*

DATE	DELIVERABLE
Oct 16	End of reading week: RTL for the PFD, prescaler, divider and lock detector, CocoTB passing; first charge-pump and VCO schematics.
Nov 5	Evaluation: each stage simulated in ngspice; the full loop locks in simulation at the typical corner.
Nov 26	System verification: layout DRC and LVS clean, Tiny Tapeout precheck green.
Dec 3	Documentation and a silicon test plan.

The work is split by loop stage, so each person owns blocks from schematic to layout. Ihsan takes detect and filter: the PFD, charge pump and loop filter. Dhyey takes oscillate and divide: the VCO, prescaler, $\div N$ divider and lock detector. Both do top-level integration and full-loop simulation, documentation, GitHub and the presentations.

Pins from slide 2 and Table 4 of the proposal; milestones and roles from slide 3 (commit 0aeb5c8, 1 October 2026).